Random bounded-degree regular graphs are robustly self-ordered with high probability

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Allowed standard prerequisites (editorial).

- Bounded-degree expanders, graph isomorphisms, Chernoff bounds and good error-correcting codes. Source-local colored-to-uncolored transformation and complete three-step construction included; standard cited expander/coding results not recursively reproved

1 Introduction

For a (labeled) graph $G = (V, E)$, and a bijection $\phi : V \to V'$, we denote by $\phi(G)$ the graph $G' = (V', E')$ such that $E' = \{\{\phi(u), \phi(v)\} : \{u, v\} \in E\}$, and say that G' is isomorphic to G. The set of automorphisms of the graph $G = (V, E)$, denoted $\mathtt{aut}(G)$, is the set of permutations that preserve the graph G; that is, $\pi \in \mathtt{aut}(G)$ if and only if $\pi(G) = G$. We say that a graph is asymmetric (equiv., self-ordered) if its set of automorphisms is a singleton, which consists of the trivial automorphism (i.e., the identity permutation). We actually prefer the term *self-ordered*, because we take the perspective that is offered by the following equivalent definition.

Definition 1.1 (Self-ordered (a.k.a asymmetric) graphs)**.** The graph $G = ([n], E)$ is self-ordered if for every graph $G' = (V', E')$ that is isomorphic to G there exists a unique bijection $\phi : V' \to [n]$ such that $\phi(G') = G$.

In other words, given an isomorphic copy $G' = (V', E')$ of a fixed graph $G = ([n], E)$, there is a unique bijection $\phi : V' \to [n]$ that orders the vertices of G' such that the resulting graph (i.e., $\phi(G')$) is identical to G. Indeed, if $G' = G$, then this unique bijection is the identity permutation. [1]

[1] Naturally, we are interested in efficient algorithms that find this unique ordering, whenever it exists; such algorithms are known when the degree of the graph is bounded [5].

In this work, we consider a feature, which we call *robust self-ordering*, that is a quantitative version self-ordering. Loosely speaking, a graph $G = ([n], E)$ is robustly self-ordered if, for every permutation $\pi : [n] \to [n]$, the size of the symmetric difference between G and $\pi(G)$ is proportional to the number of non-fixed-points under π; that is, $|E \triangle \{\{\pi(u), \pi(v)\} : \{u, v\} \in E\}|$ is proportional to $|\{i \in [n] : \pi(i) \neq i\}|$. (In contrast, self-ordering only means that the size of the symmetric difference is positive if the number of non-fixed-points is positive.)

Definition 1.2 (Robustly self-ordered graphs)**.** A graph $G = (V, E)$ is said to be γ-robustly self-ordered if for every permutation $\pi : V \to V$ it holds that

$$\left|E \triangle \{\{\pi(u), \pi(v)\} : \{u, v\} \in E\}\right| \;\geq\; \gamma \cdot |\{i \in [n] : \pi(i) \neq i\}|, \tag{1}$$

where $\triangle$ denotes the symmetric difference operation. An infinite family of graphs $\{G_n = ([n], E_n)\}_{n \in \mathbb{N}}$ *(such that each G_n has maximum degree d)* is called robustly self-ordered if there exists a constant $\gamma > 0$, called the robustness parameter, such that for every n the graph G_n is γ-robustly self-ordered.

Note that $|E_n \triangle \{\{\pi(u), \pi(v)\} : \{u, v\} \in E_n\}| \leq 2d \cdot |\{i \in [n] : \pi(i) \neq i\}|$ always holds (for families of maximum degree d). The term "robust" is inspired by the property testing literature (cf. [6]), where it indicates that some "parametrized violation" is reflected proportionally in some "detection parameter".

The second part of Definition 1.2 is tailored for bounded-degree graphs, which will be our focus in Section 2–6. Nevertheless, in Sections 7–10 we consider graphs of unbounded degree and unbounded robustness parameters. In this case, for a function $\rho : \mathbb{N} \to \mathbb{R}$, we say that an infinite family of graphs $\{G_n = ([n], E_n)\}_{n \in \mathbb{N}}$ is ρ-robustly self-ordered if for every n the graph G_n is $\rho(n)$-robustly self-ordered. Naturally, in this case, the graphs must have $\Omega(\rho(n) \cdot n)$ edges. [2] In Sections 7–9 we consider the case of $\rho(n) = \Omega(n)$.

6 Random Regular Graphs are Robustly Self-Ordered

While Theorem 4.1 only asserts the existence of robustly self-ordered d-regular graphs, we next show that almost all d-regular graphs are robustly self-ordered. This extends work in probabilistic graph theory, which proves a similar result for the weaker notion of self-ordered (a.k.a asymmetric) graphs [1, 2].

Theorem 6.1 (Random d-regular graphs are robustly self-ordered)**.** *For any sufficiently large constant d, a random $2d$-regular n-vertex graph is robustly self-ordered with probability $1 - o(1)$.*

Recall that, with very high probability, these graphs are expanders. We mention that the proof of Theorem 4.1 actually established that n-vertex graphs drawn from a weird distribution (which has min-entropy $\Omega(n)$) are robustly self-ordered with probability $1 - o(1)$. However, this is established by using the edge-coloring variant, and requires employing the transformation presented in Section 2.1. In contrast, the following proof works directly with the original (uncolored) variant, and is completely self-contained.

[2] Actually, all but at most one vertex must have degree at least $\rho(n)/2$.

Proof. The proof is quite similar to the proof Claim 4.1.1, but it faces complications that were avoided in the prior proof by using edge-colors and implicitly directed edges. Specifically, for candidate permutations $\pi_1, \ldots, \pi_d : [n] \to [n]$ (to be used in the construction) and all (non-trivial) permutations $\mu : [n] \to [n]$, the proof of Claim 4.1.1 considered events of the form $(\forall j \in [d])\ \pi_j(i) = \mu(\pi_j(\mu^{-1}(i)))$, whereas here we shall consider events of the form $\{\pi_j^b(i) : j \in [d]\ \&\ b \in \{\pm 1\}\} = \{\mu(\pi_k^c(\mu^{-1}(i))) : k \in [d]\ \&\ c \in \{\pm 1\}\}$. That is, rather than considering a sequence of equations that refers to single π_j's, we consider a sequence of equations that refer to the set of all π_j's, which were expressed above in terms of equations between multi-sets. These multi-set equations will be reduced to equations among sequences by considering all possible ordering of these multi-sets. This amounts to taking a union bound over all possible ordering and results in a more complicated analysis (due to the π_j^{-1}'s) and much more cumbersome notation.

To facilitate the proof, we use the standard methodology (cf. [3, Apdx. 2]) of first proving the result in the *random permutation model*, then transporting it to the *configuration model* (by using a general result of [4]), and finally conditioning on the event that the generated graph is simple (which occurs with positive constant probability). Indeed, both models generate multi-graphs that are not necessarily simple graphs (i.e., these multi-graphs may have self-loops and parallel edges). We also use the fact that the simple graphs that are generated by the configuration model (for degree d') are uniformly distributed among all d'-regular graphs.

Recall that in the random permutation model a $2d$-regular n-vertex multi-graph is generated by selecting uniformly and independently d permutations $\pi_1, \ldots, \pi_d : [n] \to [n]$. The multi-graph, denoted $G_{(\pi_1,\ldots,\pi_d)}$, consists of the edge multi-set $\bigcup_{j\in[d]}\{\{i, \pi_j(i)\} : i \in [n]\}$, where the $2j^{\text{th}}$ (resp., $(2j-1)^{\text{st}}$) neighbor of vertex i is $\pi_j(i)$ (resp., $\pi_j^{-1}(i)$). Note that this multi-graph may have self-loops (due to $\pi_j(i) = i$), which contributed two units to the degree of a vertex, as well as parallel edges (due to $\pi_j(i) = \pi_k(i)$ for $j \neq k$ and $\pi_j(i) = \pi_k^{-1}(i)$ for any j, k). We denote the jth neighbor of vertex i by $g_j(i)$; that is, $g_j(i) = \pi_{j/2}(i)$ if j is even, and $g_j(i) = \pi^{-1}_{(j+1)/2}(i)$ otherwise.

Consider an arbitrary permutation $\mu : [n] \to [n]$, and let $T = \{i \in [n] : \mu(i) \neq i\}$ be its set of non-fixed-point. We shall show that, with probability $1 - \exp(-\Omega(d \cdot |T| \cdot \log n))$ over the choice of $\overline{\pi} = (\pi_1, \ldots, \pi_d)$, the size of the symmetric difference between $G_{\overline{\pi}}$ and $\mu(G_{\overline{\pi}})$ is $\Omega(|T|)$. Note that this difference is (half) the sum over $i \in [n]$ of the size of the symmetric difference between the multi-set of neighbors of vertex i in $G_{\overline{\pi}}$ and the multi-set of neighbors of vertex i in $\mu(G_{\overline{\pi}})$. We refer to the latter difference by the phrase *the contribution of vertex i to the difference between $G_{\overline{\pi}}$ and $\mu(G_{\overline{\pi}})$*.

As a warm-up, we first show that each element of T contributes a non-zero number of units to the difference (between $G_{\overline{\pi}}$ and $\mu(G_{\overline{\pi}})$) with probability $1 - O(\text{poly}(d)/n)^{d/6}$ over the choice of $\overline{\pi}$. Recalling that $i \in T$ contributes to the difference (between $G_{\overline{\pi}}$ and $\mu(G_{\overline{\pi}})$) if the multi-sets of its neighbors in $G_{\overline{\pi}}$ and $\mu(G_{\overline{\pi}})$ differ, it follows that $i \in T$ contributes to the difference if *for every permutation $\sigma : [2d] \to [2d]$ there exists $j \in [2d]$ such that $g_j(i) \neq \mu(g_{\sigma(j)}(\mu^{-1}(i)))$*. Hence, the complementary event holds (i.e., i does not contribute to the difference) if and only if there exists permutation $\sigma : [2d] \to [2d]$ such that for every $j \in [2d]$ it holds that $g_j(i) = \mu(g_{\sigma(j)}(\mu^{-1}(i)))$. Thus, the probability that i does

not contribute to the difference is given by

$$\begin{aligned}&\Pr_{\overline{\pi}}\left[\exists\sigma\in\mathrm{Sym}_{2d}\ (\forall j\in[2d])\ \ g_j(i)=\mu(g_{\sigma(j)}(\mu^{-1}(i)))\right]\\ &\quad\leq\quad (2d)!\cdot\max_{\sigma\in\mathrm{Sym}_{2d}}\left\{\Pr_{\overline{\pi}}\left[(\forall j\in[2d])\ \ g_j(i)=\mu(g_{\sigma(j)}(\mu^{-1}(i)))\right]\right\}.\end{aligned}\tag{2}$$

Fixing σ that maximizes the probability, and denoting it σ_i, consider any $J_i\subseteq[d]$ such that for the j's in J_i the multi-sets $\{j,\lceil\sigma_i(2j)/2\rceil\}$'s are disjoint (i.e., $\{j,\lceil\sigma_i(2j)/2\rceil\}\cap\{k,\lceil\sigma_i(2k)/2\rceil\}=\emptyset$ for any $j\neq k\in J_i$). Note that we may select J_i such that $|J_i|\geq d/6$, since taking j to J_i rules out taking (to J_i) at most five other values (i.e., letting $v\stackrel{\text{def}}{=}\lceil\sigma_i(2j)/2\rceil$, the value k is ruled out if either $k\in\{j,v\}$ or $\sigma_i(2k)\in\{2j-1,2j,2v-1,2v\}$).

Using this feature of J_i, we prove –

Claim 6.1.1. *(Warm-up):*[3] (2) is upper-bounded by $(2d)^{2d}\cdot(3/n)^{|J_i|}$.

Proof. Recalling that σ_i maximizes (2), we upper-bound (2) by

$$\begin{aligned}&(2d)!\cdot\Pr_{\overline{\pi}}\left[(\forall j\in J_i)\ \ g_{2j}(i)=\mu(g_{\sigma_i(2j)}(\mu^{-1}(i)))\right]\\ &\quad=\quad (2d)!\cdot\prod_{j\in J_i}\Pr_{\pi_j,\pi_{\lceil\sigma_i(2j)/2\rceil}}\left[g_{2j}(i)=\mu(g_{\sigma_i(2j)}(\mu^{-1}(i)))\right]\end{aligned}\tag{3}$$

where the equality uses the disjointness of the multi-sets $\{j,\lceil\sigma_i(2j)/2\rceil\}$ for the j's in J_i. Next, letting $\sigma'_i(2j)\stackrel{\text{def}}{=}\lceil\sigma_i(2j)/2\rceil$, and $\sigma''_i(2j)\stackrel{\text{def}}{=}(-1)^{\sigma_i(2j)\bmod 2}$ (so that $\sigma_i(2j)=2\sigma'_i(2j)-0.5(1-\sigma''_i(2j))$), we upper-bound (3) by

$$(2d)!\cdot\prod_{j\in J_i}\Pr_{\pi_j,\pi_{\sigma'_i(2j)}}\left[\pi_j(i)=\mu(\pi^{\sigma''_i(2j)}_{\sigma'_i(2j)}(\mu^{-1}(i)))\right]\ <\ (2d)^{2d}\cdot(3/n)^{|J_i|},\tag{4}$$

where $\Pr_{\pi_j,\pi_j}[\cdot]$ stands for $\Pr_{\pi_j}[\cdot]$ and π^1 stands for π, while the inequality is justified by considering the following three cases (w.r.t each $j\in J_i$).

1. If $k\stackrel{\text{def}}{=}\sigma'_i(2j)\neq j$ (equiv., $\sigma_i(2j)\notin\{2j-1,2j\}$), then, letting $b=\sigma''_i(2j)$, the corresponding factor in the l.h.s of (4) is

   $$\Pr_{\pi_j,\pi_k}\left[\pi_j(i)=\mu(\pi^b_k(\mu^{-1}(i)))\right]$$

   which equals $1/n$ by fixing π_k, letting $v=\mu(\pi^b_k(\mu^{-1}(i)))$, and using $\Pr_{\pi_j}[\pi_j(i)=v]=1/n$.

2. If $\sigma_i(2j)=2j$, then the corresponding factor in the l.h.s of (4) is

   $$\Pr_{\pi_j}\left[\pi_j(i)=\mu(\pi_j(\mu^{-1}(i)))\right]$$

   which is at most $1/(n-1)$ since $\mu(i)\neq i$; specifically, fixing the value of $\pi_j(\mu^{-1}(i))$, and denoting this value by v, leaves $\pi_j(i)$ uniformly distributed in $[n]\setminus\{v\}$, which means that $\Pr_{\pi_j}[\pi_j(i)=\mu(v)\,|\,v=\pi_j(\mu^{-1}(i))]\leq 1/(n-1)$ (where equality holds if $\mu(v)\neq v$).

[3] One may obtain a better bound of $O(d/n)^{2d}$ by analyzing (2) directly, by considering all the $2d$ events and accounting for their small dependency. On the other hand, we can obtain higher robustness parameter by considering smaller sets J_i's (say of size $d/12$), which suffice for counting vertices that contribute (say) $d/2$ units to the difference between $G_{\overline{\pi}}$ and $\mu(G_{\overline{\pi}})$.

3. If $\sigma_i(2j) = 2j-1$, then the corresponding factor in the l.h.s of (4) is

$$\Pr_{\pi_j}\left[\pi_j(i) = \mu(\pi_j^{-1}(\mu^{-1}(i)))\right]$$

which is shown to be less than $3/n$. In this case, we consider two sub-cases depending on whether or not $\pi_j(i) = \mu^{-1}(i)$, while noting that the first case occurs with probability $1/n$ whereas $\Pr_{\pi_j}[\pi_j(i) = \mu(\pi_j^{-1}(\mu^{-1}(i)))|\pi_j(i) \neq \mu^{-1}(i)] \leq 1/(n-1)$.

Hence, each of the factors in the l.h.s of (4) is upper-bounded by $3/n$, and the claim follows. □

The general case. The same argument generalizes to any set $I \subseteq T$ such that $I \cap \mu(I) = \emptyset$. In such a case we get

$$\begin{aligned}&\Pr_{\overline{\pi}}\left[(\forall i \in I)\,(\exists \sigma_i \in \mathrm{Sym}_{2d})\,(\forall j \in [2d])\;\; g_j(i) = \mu(g_{\sigma_i(j)}(\mu^{-1}(i)))\right] \\ &\quad\leq\quad (2d)!^{|I|} \cdot \max_{\sigma_1,\ldots,\sigma_n}\left\{\Pr_{\overline{\pi}}\left[(\forall i \in I)\,(\forall j \in [2d])\;\; g_j(i) = \mu(g_{\sigma_i(j)}(\mu^{-1}(i)))\right]\right\} \quad (5)\end{aligned}$$

Claim 6.1.2. *(Actual analysis):* (5) is upper-bounded by

$$(2d)^{2d\cdot|I|} \cdot (3/(n-2(|I|-1)))^{|I|\cdot d/6}. \quad (6)$$

Proof. Fixing $\sigma_1, ..., \sigma_n$ that maximize (5), we proceed as in the proof of Claim 6.1.1. For every $i \in I = \{i_1, \ldots, i_m\}$,

we fix a set J_i of size at least $d/6$ such that the multi-sets $\{j, \lceil \sigma_i(2j)/2 \rceil\}$'s are disjoint, and refer to the events $E_{j,k}(\pi_1, \ldots, \pi_{2d})$ that depend only on the value of π_j and $\pi^{\sigma''_{i_k}(2j)}_{\sigma'_{i_k}(2j)}$ on the points $i_1, \ldots, i_{k-1}$ and $\mu^{-1}(i_1), \ldots, \mu^{-1}(i_{k-1})$, respectively, where $\sigma'_i(2j) \stackrel{\text{def}}{=} \lceil \sigma_i(2j)/2 \rceil$, and $\sigma''_i(2j) \stackrel{\text{def}}{=} (-1)^{\sigma_i(2j) \bmod 2}$ (as in the proof of Claim 6.1.1). Specifically, $E_{j,k}(\pi_1, \ldots, \pi_{2d})$ is the event

$$(\forall k' \in [k-1])\;\; g_{2j}(i_{k'}) = \mu(g_{\sigma_{i_{k'}}(2j)}(\mu^{-1}(i_{k'})))$$

which can be written as

$$(\forall k' \in [k-1])\;\; \pi_j(i_{k'}) = \mu(\pi^{\sigma''_{i_{k'}}(2j)}_{\sigma'_{i_{k'}}(2j)}(\mu^{-1}(i_{k'}))).$$

Using the disjointness of the $\{j, \lceil \sigma_i(2j)/2 \rceil\}$'s in J_i, we upper-bound (5) by

$$\begin{aligned}&(2d)!^m \cdot \prod_{k\in[m]}\prod_{j\in J_{i_k}} \Pr_{\pi_1,\ldots,\pi_{2d}}\left[g_{2j}(i_k) = \mu(g_{\sigma_{i_k}(2j)}(\mu^{-1}(i_k)))\,|E_{j,k}(\pi_1, ..., \pi_{2d})\right] \\ &=\quad (2d)!^m \cdot \prod_{k\in[m]}\prod_{j\in J_{i_k}} \Pr_{\pi_1,\ldots,\pi_{2d}}\left[\pi_j(i_k) = \mu(\pi^{\sigma''_{i_k}(2j)}_{\sigma'_{i_k}(2j)}(\mu^{-1}(i_k)))\,|E_{j,k}(\pi_1, ..., \pi_{2d})\right] \quad (7)\end{aligned}$$

Now, when analyzing the foregoing conditional probability in (7), we consider two cases for each $k \in [m]$ and $j \in J_{i_k}$. If $j \neq \sigma'_{i_k}(2j)$, then we fix the value of each of these two permutations (i.e., π_j and $\pi_{\sigma'_{i_k}(2j)}$) on the corresponding $k-1$ points that occur in the condition $E_{j,k}$, and the value of these permutations

on the kth points (i.e., i_k and $\mu^{-1}(i_k)$) is restricted accordingly (i.e., to the remaining $n-(k-1)$ values). Otherwise (i.e., $j=\sigma'_{i_k}(2j)$), we fix the value of π_j on these $2(k-1)$ points. Hence, the argument in the warm-up analysis applies with n replaces by either $n-(k-1)$ or $n-2(k-1)$. It follows that (7) is upper-bounded by

$$(2d)!^m \cdot \prod_{k\in[m]} (3/(n-2(m-1)))^{|J_{i_k}|}.$$

Using $|J_{i_k}| \geq d/6$ for every $k \in [m]$,
the claim follows. □

Recall that (6) refers to a fixed set $I \subseteq T$ such that $I \cap \mu(I) = \emptyset$, and that it constitutes an upper bound on the probability (over the choice of $\overline{\pi}$) that, for each $i \in I$ there exists a permutation $\sigma_i : [2d] \to [2d]$ such that $g_j(i) = \mu(g_{\sigma_i(j)}(\mu^{-1}(i)))$ holds for all $j \in [2d]$. This upper bound (i.e., $(2d)^{2d\cdot|I|} \cdot (3/(n-2(|I|-1)))^{|I|\cdot d/6}$) simplifies to $(2d)^{2d\cdot|I|} \cdot (5/n)^{|I|\cdot d/6}$, provided that $|I| \leq \lceil n/6 \rceil$.

Recalling that $t \stackrel{\text{def}}{=} |T| \in [n]$, we shall upper-bound the probability (over the choice of $\overline{\pi}$) that T contains a $\lceil t/2 \rceil$-subset T' such that for each $i \in T'$ there exists a permutation $\sigma_i : [2d] \to [2d]$ such that $g_j(i) = \mu(g_{\sigma_i(j)}(\mu^{-1}(i)))$ holds for all $j \in [2d]$. We do so by taking a union bound over all $\lceil t/6 \rceil$-subsets I such that $I \cap \mu(I) = \emptyset$ and for each $i \in I$ there exists a permutation $\sigma_i : [2d] \to [2d]$ such that $g_j(i) = \mu(g_{\sigma_i(j)}(\mu^{-1}(i)))$ holds for all $j \in [2d]$. (Note that such a $\lceil t/6 \rceil$-subset I exists in each $\lceil t/2 \rceil$-subset T', and that $\lceil t/6 \rceil < n/3$.) Using the aforementioned simplified form of (6), we conclude that, with probability at most

$$\binom{t}{\lceil t/6 \rceil} \cdot (2d)^{2d\cdot\lceil t/6 \rceil} \cdot (5/n)^{\lceil t/6 \rceil \cdot d/6} \;<\; 2^t \cdot (5\cdot(2d)^{12}/n)^{\lceil t/6 \rceil \cdot d/6} \;=\; \exp(-\Omega(dt \log n))$$

over the choice of $\overline{\pi}$, the set T contains no $\lceil t/6 \rceil$-subset I as above. This means that, with probability at most $\exp(-\Omega(dt \log n))$, less than $t/2$ of the indices $i \in T$ contribute a non-zero number of units to the difference (between $G_{\overline{\pi}}$ and $\mu(G_{\overline{\pi}})$).

Letting $c' = 1/2$ and considering all (non-trivial) permutations $\mu : [n] \to [n]$, we conclude that the probability, over the choice of $\overline{\pi}$, that $G_{\overline{\pi}}$ is not c'-robustly self-ordered is at most

$$\begin{aligned}\sum_{t\in[n]} \binom{n}{t} \cdot \exp(-\Omega(dt \log n)) &= \sum_{t\in[n]} \exp(-\Omega((d-O(1)) \cdot t \log n)) \\ &= \exp(-\Omega((d-O(1)) \cdot \log n)),\end{aligned}$$

and the claim follows for the permutation model (and for any sufficiently large d).

As stated upfront, using the general result of [4, Thm. 1.3], we infer that a uniformly distributed $2d$-regular n-vertex multi-graph fails to be c'-robustly self-ordered with probability $o(1)$. Lastly, recalling that such a $2d$-regular multi-graph is actually a simple graph with probability $\exp(-((2d)^2-1)/4)$, the theorem follows. □

Digest. The proof of Theorem 6.1 is quite similar to the proof Claim 4.1.1, but it faces two complications that were avoided in the prior proof (by using edge-colors and implicitly directed edges). Most importantly, the current proof has to handle equality between multi-sets instead of equality between sequences. This is done by considering all possible ordering of these multi-sets, which amounts to taking a union bound over all possible ordering and results in more complicated analysis and notation. (Specifically, see the introduction of σ_i's and J_i's and the three cases analyzed in the warm-up.) In addition, since edges are defined by permutations over the vertex-set rather than by perfect matching, we have to consider both the forward and backward direction of each permutation, which results in further complicating the analysis and the notation. (Specifically, see the introduction of σ'_i's and σ''_i's and the three cases analyzed in the warm-up.)

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